Code No: 135CH **R16**

JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY HYDERABAD
B. Tech III Year I Semester Examinations, March - 2021
FUNDAMENTALS OF MECHANICAL ENGINEERING
(Common to CE, EEE, CSE, MIE)

Time: 3 hours **Max. Marks: 75**

Answer any five questions
All questions carry equal marks
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1. Explain the following:
   a) Fossil fuels
   b) Nuclear fuels. [7+8]

2.a) What are the limitations of First law of thermodynamics?
   b) Compare the first law of thermodynamics with its corollaries. [7+8]

3. At 1.2 MPa, 250^0C steam enters into a turbine and expands to 30^0C. Determine the work output of turbine for 10 kg/s flow rate steam. [15]

4. Explain the function of fusible plug, and where is it located in a Babcock and Wilcox boiler? [15]

5. An ideal diesel cycle operates on 1 kg of standard air with an initial pressure of 0.98 bar and a temperature of 35^0C. The pressure at the end of compression is 33 bar and the cut-off is 6% of the stroke. Determine: (a) compression ratio, (b) percentage clearance, and (c) heat supplied. [5+5+5]

6. During testing of single cylinder two-stroke petrol engine, following data were obtained—brake torque = 640 Nm; cylinder diameter = 21 cm, speed = 250 rpm; stroke = 28 cm; m.e.p. = 5.6 bar; oil consumption = 8.16 kg/hr; C.V. = 42,705 *kJ/kg*. Determine: (a) mechanical efficiency, (b) indicated thermal efficiency, and (c) Brake specific fuel consumption. [5+5+5]

7.a) What is multistage compression? What are the advantages of it?
   b) Derive the expression for work done in multistage compression. [7+8]

8. Explain standard vapor compression refrigeration cycle with T-S and P-H diagram. What is the effect of sub-cooling on the performance of vapor compression refrigeration system? [15]

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